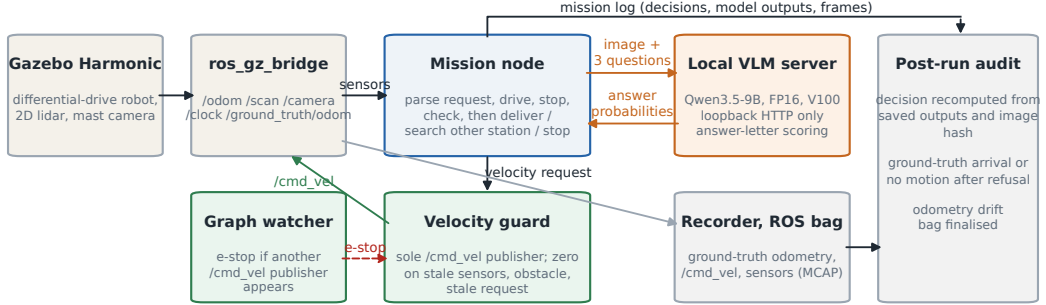
mission log (decisions, model outputs, frames)
Gazebo Harmonic
differential-drive robot, 2D lidar, mast camera
ros_gz_bridge
/odom /scan /camera /clock /ground_truth/odom
sensors
Mission node
parse request, drive, stop, check, then deliver / search other station / stop
image + 3 questions
answer probabilities
Local VLM server
Qwen3.5-9B, FP16, V100 loopback HTTP only answer-letter scoring
Post-run audit
decision recomputed from saved outputs and image hash
ground-truth arrival or no motion after refusal
odometry drift bag finalised
velocity request
/cmd_vel
Graph watcher
e-stop if another /cmd_vel publisher appears
e-stop
Velocity guard
sole /cmd_vel publisher; zero on stale sensors, obstacle, stale request
Recorder, ROS bag
ground-truth odometry, /cmd_vel, sensors (MCAP)